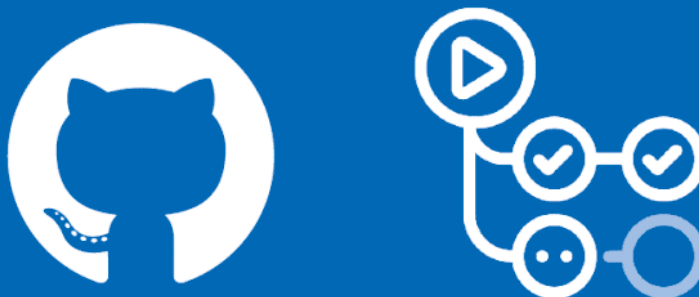


Continuous Integration with GitHub Actions

Here are a few insights into the core principles of DevOps and some practical guidance on using GitHub Actions.



DevOps isn't the name of a team or a job title or a particular technology. Instead, it's a set of processes, ideas, and techniques that combines people, cultural practices, and technologies. The goal is to make software delivery vastly more efficient. DevOps helps teams ship high-quality products faster by reducing the friction between writing, testing, and deploying code.

What is the structure of a DevOps model? Well, it does not work on a particular model; rather it is an approach to building software to enhance the entire development cycle. DevOps fundamentally seeks to transform organisations by bringing traditionally siloed teams together

across every part of the software development life cycle.

Key aspects of DevOps

Collaboration: DevOps needs close cooperation between operations and development for a productive environment.

Version control: Version control is an integral part of DevOps. A version control system is designed to automatically record file changes and preserve records of previous file versions.

Automation: It establishes feedback loops between those in charge of maintaining the underlying infrastructure and those in charge of developing the core software.

Incremental releases: Successful DevOps practices rely on incremental releases, which are characterised by the speedy delivery of feature additions and updates. Incremental releases enable development teams to swiftly respond to business needs.

Orchestration: Orchestration aids in configuring, managing, and coordinating the infrastructure necessary for an application to function properly.

Pipeline: The term 'pipeline' frequently arises in discussions on DevOps. A DevOps pipeline sequences various stages of the software development life cycle like compiling and analysing the code, unit testing, building the image, containerising the image, and more.